

Stereovision-Based Estimation of Relative Dynamics Between Noncooperative Satellites: Theory and Experiments

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Abstract—Estimating the relative pose and motion of cooperative satellites using on-board sensors is a challenging problem. When the satellites are noncooperative, the problem becomes even more complicated, as there might be poor *a priori* information about the motion and structure of the target satellite. In this paper, the mentioned problem is solved by using only visual sensors, which measurements are processed through robust filtering algorithms. Using two cameras mounted on a chaser satellite, the relative state with respect to a target satellite, including the position, attitude, and rotational and translational velocities, is estimated. The new approach employs a stereoscopic vision system for tracking a set of feature points on the target spacecraft. The perspective projection of these points on the two cameras constitutes the observation model of an iterated extended Kalman filter (IEKF) estimation scheme. Using new theoretical results, the information contained in the visual data is quantified using the Fisher information matrix. It is shown that, even in the noncooperative case, there is information that can be extracted pertaining to the relative attitude and target structure. Finally, a method is proposed for rendering the relative motion filtering algorithm robust to uncertainties in the target's inertia tensor. This is accomplished by endowing the IEKF with a maximum *a posteriori* identification scheme for determining the most probable inertia tensor from several available hypotheses. The performance of the new filtering algorithm is validated by Monte-Carlo simulations. Also a preliminary experimental evaluation is provided.

Index Terms—Estimation theory, satellite dynamics, stereovision, tracking.

I. INTRODUCTION

ESTIMATING the relative state between space vehicles has attracted considerable attention in the last decade [1]. Frequently, the relative state estimation problem is treated in spacecraft formation flying [2] or rendezvous and docking [3], which are considered to be cooperative space missions. In these cases, the relative sensing technique relies on information shared between the spacecraft. Normally, the relative sensor comprises of two separate devices such as two

differential global positioning system (GPS) receivers [4] or position-sensing diodes and light beacons [2].

However, some spacecraft missions may involve interactions between noncooperative vehicles. These missions include tracking of resident space objects and on-orbit replenishment. When the satellites are noncooperative, the problem of relative state estimation becomes far more complicated, as there might be poor *a priori* information about the motion and structure of the target satellite. Thus, the cooperative estimation architecture is inadequate.

Noncooperative formation flying can be realized by exploiting recent developments in the field of vision-based motion reconstruction. Resorting to vision-based estimation benefits from the passive nature of visual sensory. In contrary to other methods, this setting does not require that unique indicators be placed on the target object. Therefore, the passive vision-based approach seems to be appealing for noncooperative applications.

In single-camera vision-based systems, the 2-D projections of the 3-D scene are acquired by a body-mounted camera. This leads to nonlinear implicit measurement equations. In addition, the magnitude of the translational velocity can be recovered only up to scale, yielding partial state observability. However, in this paper, it is shown that the entire relative state (pose and motion) can be estimated using two cameras instead of a single one. Through utilizing stereoscopic vision, depth information can be recovered and used for reconstructing the magnitude of the translational velocity vector in addition to its direction.

Stereovision was used in the past for estimating the relative state of two cooperative spacecraft. The algorithm in [5], for example, utilizes a sequence of stereo images for determining the relative position and attitude of an orbiting satellite. Nevertheless, its capabilities in the noncooperative settings are limited, as it assumes that the feature point locations on the target are known. If some *a priori* information about the target geometry is available, model matching techniques [6] can be used together with stereovision.

There are few papers in the literature that address the problem of relative state estimation with respect to noncooperative targets. In [7] and [8], the pose, motion, and structure of an orbiting object are estimated using several cooperative satellites equipped with 3-D vision sensors. The main drawback of this method, however, is that it involves the acquisition of thousands of uniformly distributed data points for accurately tracking the rotating structure. In addition,

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3-D sensors such as laser radars are active devices, whereas a stereo-rig provides 3-D information using passive measurements. Thus, using two cameras instead of a single 3-D sensor provides lower signatures, consumes less energy, and is usually more cost effective than 3-D sensors. Xu *et al.* [9] develop an interesting approach in which stereovision is used for autonomous rendezvous and robotic capturing of noncooperative targets. However, it does not provide guidelines for practical implementation of the newly-derived concepts.

As claimed in some of the aforementioned papers, the scaling factor ambiguity associated with stereovision can be resolved assuming the baseline separating the two cameras is known. The feature points can be then accurately tracked by measuring their projections on the right and left stereo rig planes. In addition, the target attitude can be obtained by observing several points simultaneously [10].

Estimating the relative state directly from the visual measurements involves the formulation of highly nonlinear observation and process models. These normally consist of additional uncertainties associated with noncooperative targets. The main purpose of this paper, therefore, is to design a robust filtering scheme for handling the additional complications encountered in such scenarios. The estimation scheme proposed herein consists of multiple iterated extended Kalman filters (IEKFs) and a maximum *a posteriori* (MAP) inertia tensor identification algorithm. This approach is based on the same rationale of the acclaimed multiple-model KF technique [11], which is widely used for filtering under modeling uncertainties. The key idea underlying this architecture can be summarized as follows. Filtering inconsistencies, which are normally attributed to the EKF working in extreme conditions (in which the underlying assumptions of Gaussian processes and small estimation errors are violated) can be considerably mitigated by propagating a conditional probability density function (PDF) rather than a joint PDF. Therefore, each individual IEKF approximates the PDF of the relative state conditioned on the inertia tensor. This, in turn, facilitates the application of a Bayesian MAP estimator for recognizing the most probable inertia hypothesis based on the obtained conditional.

Therefore, compared with previous papers (e.g., [12]), this paper offers a complete theoretical and a preliminary experimental development of stereovision-based estimation of the relative state between noncooperative satellites. The uncertainty associated with the target inertia and feature point locations is mitigated based on robust filtering.

II. MATHEMATICAL FORMULATION

Assume that two spacecraft are in orbit around the earth. It is assumed that the chaser spacecraft, denoted by C , is equipped with vision sensors used for acquiring images of a set of N feature points located on a target spacecraft, T . These feature points can be represented as N vectors, $\{\mathbf{P}_i\}_{i=1}^N$, in some given reference frame.

The reference frames and notations used throughout this paper are similar to those in [13] and are briefly repeated here for completeness: $\mathcal{L}$ is a local-vertical, local-horizontal (LVLH) Cartesian reference frame fixed to the chaser spacecraft center-of-mass (CM), with $\hat{\mathbf{x}}$ being a unit vector directed

from the spacecraft radially outward, $\hat{\mathbf{z}}$ is normal to the chaser's orbital plane, positive in the direction of the instantaneous orbital angular momentum vector, and $\hat{\mathbf{y}}$ completes the dextral triad; $\mathcal{L}$ is a Cartesian dextral body-fixed reference frame attached to the chaser spacecraft's CM; and $\mathcal{T}$ is a Cartesian dextral body-fixed reference frame attached to the target spacecraft's CM. In the following, we assume that the body-fixed frame $\mathcal{L}$ is aligned with the orbital frame $\mathcal{L}$, thus the satellite is three-axis stabilized about its LVLH frame.

Let $\mathbf{P}_i \in \mathbb{R}^3$ be an arbitrary feature vector directed from the origin of the coordinate system $\mathcal{T}$ to some feature point P_i on T . The vector $\boldsymbol{\rho}_i = [\rho_i^{(1)}, \rho_i^{(2)}, \rho_i^{(3)}]^T \in \mathbb{R}^3$ denotes the position of P_i relative to the CM of C , resolved in frame $\mathcal{L}$, and $\dot{\boldsymbol{\rho}}_i = [\dot{\rho}_i^{(1)}, \dot{\rho}_i^{(2)}, \dot{\rho}_i^{(3)}]^T$ denotes the corresponding components of the relative translational velocities resolved in $\mathcal{L}$. A feature point located on the CM of T will be denoted by P_0 , and the corresponding relative position and velocity vectors will be denoted by $\boldsymbol{\rho}_0$ and $\dot{\boldsymbol{\rho}}_0$, respectively.

The attitude of T relative to C is parameterized using the rotation quaternion $\mathbf{q} = [\boldsymbol{\varrho}^T, q_4]^T \in \mathbb{S}^3$ where $\boldsymbol{\varrho}$ and q_4 denote its vector and scalar parts, respectively, and $\mathbb{S}^3$ is the unit 3-sphere. The attitude matrix associated with $\mathbf{q}$, denoted by $D(\mathbf{q})$, is given by [14]

$$D(\mathbf{q}) = [(q_4)^2 - \boldsymbol{\varrho}^T \boldsymbol{\varrho}] \mathbb{I}_{3 \times 3} + 2\boldsymbol{\varrho} \boldsymbol{\varrho}^T - 2q_4[\boldsymbol{\varrho} \times] \quad (1)$$

where $\mathbb{I}$ is the identity matrix and $[\mathbf{v} \times]$ is the cross-product equivalent matrix (i.e., the matrix satisfying $[\mathbf{v} \times] \mathbf{u} = \mathbf{v} \times \mathbf{u}$). In addition, we denote by $\boldsymbol{\omega} \in \mathbb{R}^3$ the relative angular velocity of T with respect to C , expressed in the chaser body frame, thus

$$\boldsymbol{\omega} = D(\mathbf{q})\boldsymbol{\omega}_T - \boldsymbol{\omega}_C. \quad (2)$$

A. Problem Statement

Our objective can be summarized as follows: Using a sequence of stereo images, estimate the relative state

$$\mathbf{x} = [\boldsymbol{\rho}_0^T, \dot{\boldsymbol{\rho}}_0^T, \boldsymbol{\omega}^T, \mathbf{q}^T, \mathbf{P}_1^T, \dots, \mathbf{P}_N^T]^T \in \mathbb{R}^{13+3N} \quad (3)$$

and other unknown parameters (e.g., the inertia tensor). Formally, $\mathbf{x}$ is assumed to be a random process for which the evolution is described by means of a stochastic differential equation. Following the common engineering notation, we may write:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \boldsymbol{\xi}(t) \quad (4)$$

where $\boldsymbol{\xi}(t)$ is a zero-mean white noise with power spectral density matrix $E[\boldsymbol{\xi}(t)\boldsymbol{\xi}(s)^T] = Q\delta(t-s)$. The explicit form of the nonlinear vector-valued function $\mathbf{f}$ will be elaborated next.

B. Dynamical Model

The translational relative dynamics between the CMs of T and C , resolved in $\mathcal{L}$, can be written as shown in (5) at the top of the next page [13], where v_C is the chaser's true anomaly and r_C is the chaser's CM distance from the attracting center.¹ The relative rotational dynamics are written for both

¹The 1-sigma accuracy of measuring the absolute distance r_C using a space-qualified GPS receiver is about 5 m. This means that the relative radial position component $\rho_0^{(1)}$ should be above this value.

$$\ddot{\mathbf{p}}_0 = \begin{bmatrix} \ddot{p}_0^{(1)} \\ \ddot{p}_0^{(2)} \\ \ddot{p}_0^{(3)} \end{bmatrix} = \begin{bmatrix} 2\dot{v}_C\dot{\rho}_0^{(2)} + \ddot{v}_C\rho_0^{(2)} + \dot{v}_C^2\rho_0^{(1)} - \frac{\mu(r_C + \rho_0^{(1)})}{\left[(r_C + \rho_0^{(1)})^2 + (\rho_0^{(2)})^2 + (\rho_0^{(3)})^2\right]^{3/2}} + \frac{\mu}{r_C^2} \\ 2\dot{v}_C\dot{\rho}_0^{(1)} - \ddot{v}_C\rho_0^{(1)} + \dot{v}_C^2\rho_0^{(2)} - \frac{\mu\rho_0^{(2)}}{\left[(r_C + \rho_0^{(1)})^2 + (\rho_0^{(2)})^2 + (\rho_0^{(3)})^2\right]^{3/2}} \\ - \frac{\mu\rho_0^{(3)}}{\left[(r_C + \rho_0^{(1)})^2 + (\rho_0^{(2)})^2 + (\rho_0^{(3)})^2\right]^{3/2}} \end{bmatrix} \quad (5)$$

spacecraft using their respective inertia tensors $I_C, I_T \in \mathbb{R}^{3 \times 3}$, and external torques, $\mathbf{N}_C, \mathbf{N}_T \in \mathbb{R}^3$

$$\begin{aligned} \dot{\boldsymbol{\omega}} = (I_C)^{-1} \{ & I_C D(\mathbf{q}) I_T^{-1} [\mathbf{N}_T - D^T(\mathbf{q})(\boldsymbol{\omega} + \boldsymbol{\omega}_C) \\ & \times I_T D^T(\mathbf{q})(\boldsymbol{\omega} + \boldsymbol{\omega}_C)] \\ & - I_C \boldsymbol{\omega}_C \times \boldsymbol{\omega} - [\mathbf{N}_C - \boldsymbol{\omega}_C \times I_C \boldsymbol{\omega}_C] \}. \end{aligned} \quad (6)$$

The relative attitude kinematics are written as follows [13]:

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} q_4 \mathbb{I}_{3 \times 3} + [\boldsymbol{\varrho} \times] \\ -\boldsymbol{\varrho}^T \end{bmatrix} \boldsymbol{\omega}. \quad (7)$$

The complete translational-rotational model can now be described using (4), where

$$\mathbf{f}(\mathbf{x}) = [\dot{\boldsymbol{\rho}}^T, \ddot{\boldsymbol{\rho}}^T, \dot{\boldsymbol{\omega}}^T, \dot{\mathbf{q}}^T, \mathbf{0}_{1 \times 3N}]^T. \quad (8)$$

The initial distribution of $\mathbf{x}$ is specified by $p(\mathbf{x}_0)$. Following this, the PDF of the process trajectory up to any given (discrete) time, $\mathbf{x}_{0:k} \triangleq \{\mathbf{x}_0, \dots, \mathbf{x}_k\}$, is obtained as follows:

$$p(\mathbf{x}_{0:k}) = p(\mathbf{x}_0) \prod_{i=1}^k p(\mathbf{x}_i | \mathbf{x}_{i-1}) \quad (9)$$

where the Markov transition PDF $p(\mathbf{x}_i | \mathbf{x}_{i-1})$ is specified by both (4) and the integration interval Δt .

The model parameterized in (3)–(8) assumes that the rotational and translational components are uncoupled. We therefore refer to (3)–(8) as the uncoupled model. However, as in [13], the kinematic rotation-translation coupling affects the relative translation. This coupling can be effectively modeled by augmenting the state with the kinematics of individual feature points, given by

$$\ddot{\boldsymbol{\rho}}_i = \ddot{\boldsymbol{\rho}}_0 + \dot{\boldsymbol{\omega}} \times \boldsymbol{\rho}_i + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_i), \quad i = 1, \dots, N \quad (10)$$

where $\boldsymbol{\rho}_i$ is the line-of-sight vector to the i th feature point and $\ddot{\boldsymbol{\rho}}_0$ is defined in (5). This way, we augment the state vector $\mathbf{x}$ thus

$$\mathbf{x} = [\boldsymbol{\rho}_0^T, \dot{\boldsymbol{\rho}}_0^T, \boldsymbol{\rho}_1^T, \dot{\boldsymbol{\rho}}_1^T, \dots, \boldsymbol{\rho}_N^T, \dot{\boldsymbol{\rho}}_N^T, \boldsymbol{\omega}^T, \mathbf{q}^T, \mathbf{P}_1^T, \dots, \mathbf{P}_N^T]^T \in \mathbb{R}^{13+9N}. \quad (11)$$

The dynamics of the state vector (11) will be referred to as the coupled model. Notably, the dimension of the state vector (11) is much larger than the dimension of the state vector (3), a fact that bares important computational implications.

Equation (5) constitutes a nonlinear relative translation model. This is preferred over linearized relative motion models

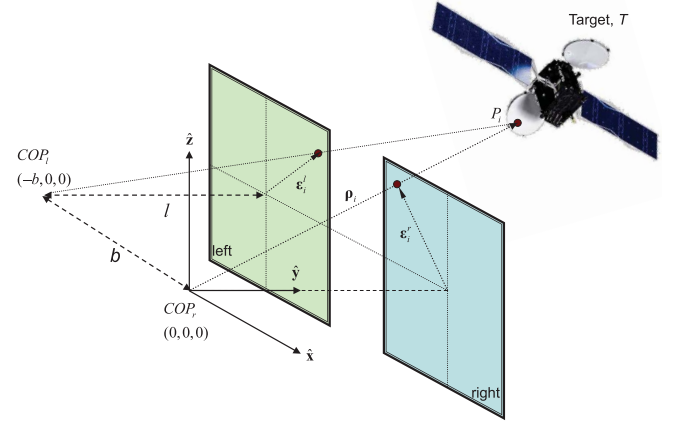


Fig. 1. Stereo rig system. The right camera's COP coincides with the origin of the chaser's body frame.

(see [1] for a thorough discussion of such models) because of two reasons: First, linearized models may yield poor relative state estimation when the distances increase to a few kilometers (and the stereoscopic system is adjusted accordingly); second, the uncoupled model is inherently nonlinear because of the coupling between rotation and translation, and hence linearization per se cannot be performed [13].

C. Observation Model

Assume that a stereo rig is mounted rigidly on C and is pointed at T , as shown in Fig. 1. The stereo rig is composed of two parallel cameras with focal length l , wherein the right camera's center-of-projection, COP_r , is located on the CM of C , and the left camera's center-of-projection, COP_l , is separated by a baseline distance b from COP_r .

For the right camera, we define a Cartesian dextral coordinate system originating at COP_r . The fundamental plane is perpendicular to the right camera's image plane. The unit vector $\hat{\mathbf{x}}$ is horizontal, pointing from COP_l to COP_r ; $\hat{\mathbf{z}}$ is vertical, positive upward; and $\hat{\mathbf{y}}$ completes the right-hand setup (see Fig. 1). In this frame, the COP vectors can be defined as (see Fig. 1)

$$\mathbf{COP}_r = [0, 0, 0]^T, \quad \mathbf{COP}_l = [-b, 0, 0]^T. \quad (12)$$

The line-of-sight $\boldsymbol{\rho}_i$ between the right (primary) camera's COP and the observed feature point, $\mathbf{P}_i$, satisfies

$$\boldsymbol{\rho}_i = \boldsymbol{\rho}_0 + D(\mathbf{q})\mathbf{P}_i \quad (13)$$

or, more generally

$$\rho_i - \rho_j = D(\mathbf{q}) (\mathbf{P}_i - \mathbf{P}_j) \quad \forall i \neq j. \quad (14)$$

When a feature point on T , $\{P_i\}_{i=1}^N$, is viewed by the stereo rig, the perspective projection model [15]–[17] describes its corresponding visual observations. The perspective projection $\pi(\rho_i) : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ transforms the feature points onto the image plane according to

$$\pi(\rho_i) = \left[\rho_i^{(1)} / \rho_i^{(2)}, \rho_i^{(3)} / \rho_i^{(2)} \right]^T. \quad (15)$$

As both cameras are separated by a fixed baseline $\mathbf{b} = [b, 0, 0]^T$, we may write down the observations corresponding to the i th feature point as (see Fig. 1)

$$\epsilon_i^r \triangleq \pi(\rho_i), \quad \epsilon_i^l \triangleq \pi(\rho_i + \mathbf{b}). \quad (16)$$

Further defining $\mathbf{z}_i \triangleq [(\epsilon_i^r)^T, (\epsilon_i^l)^T]^T$, and $\mathbf{h}_i(\mathbf{x}) \triangleq [\pi(\rho_i)^T, \pi(\rho_i + \mathbf{b})^T]^T$, the observation model can be rewritten in a compact fashion as follows:

$$\mathbf{z}_{i,k} = \mathbf{h}_i(\mathbf{x}_k) + \mathbf{n}_{i,k}, \quad i = 1, \dots, N \quad (17)$$

where $\{\mathbf{n}_{i,k}\}_{k>0}$ is a zero-mean white noise sequence with some PDF $p_{\mathbf{n}_i}$ and covariance R_k . Following this, it can be readily recognized that the overall dimension of the observation vector $\mathcal{Z}_k \triangleq [\mathbf{z}_{1,k}^T, \mathbf{z}_{2,k}^T, \dots, \mathbf{z}_{N,k}^T]^T$ is $4N$. In practice, the observation model (17) specifies the likelihood PDF $p(\mathcal{Z}_k | \mathbf{x}_k)$ as

$$p(\mathcal{Z}_k | \mathbf{x}_k) = \prod_{i=1}^N p(\mathbf{z}_{i,k} | \mathbf{x}_k) = \prod_{i=1}^N p_{\mathbf{n}_i}(\mathbf{z}_{i,k} - \mathbf{h}_i(\mathbf{x}_k)) \quad (18)$$

where it is implicitly assumed that $\{\mathbf{z}_{i,k}\}_{i=1}^N$ are conditionally independent given $\mathbf{x}_k$.

Finally, the posterior PDF of the state trajectory, on which any optimal inference relies, is expressed by means of the prior (9) and the likelihood (18), that is

$$p(\mathbf{x}_{0:k} | \mathcal{Z}_{1:k}) \propto p(\mathbf{x}_0) \prod_{i=1}^k p(\mathcal{Z}_i | \mathbf{x}_i) p(\mathbf{x}_i | \mathbf{x}_{i-1}) \quad (19)$$

where $\mathcal{Z}_{1:k} \triangleq \{\mathcal{Z}_1, \dots, \mathcal{Z}_k\}$.

III. ESTIMATION METHODOLOGY

The relative state estimation problem is strictly nonlinear, as conveyed by the underlying process and observation models. This implies that an optimal estimator is, in general, computationally infeasible; therefore, an approximate scheme should be adopted.

The minimum mean square error (MMSE) estimator is implicitly given by $E\{\mathbf{x}_k | \mathcal{Z}_{1:k}\}$, where the expectation is considered with respect to the filtering PDF, $p(\mathbf{x}_k | \mathcal{Z}_{1:k})$, which is a marginal of the posterior (19). A sequential scheme for propagating this PDF is provided by the acclaimed Bayesian recursion [18], [19]. Nonetheless, this recursion is, in general, analytically intractable in the nonlinear non-Gaussian case. There are few known methods for approximating the statistics of the filtering PDF, such as sequential Monte Carlo and grid-based approaches [19], and the nonlinear KF variants, namely

the unscented KF (UKF) and the aforementioned EKF [18], [20]. Whereas the former nonparametric approaches maintain an empirical approximation of the filtering PDF, which may become arbitrarily close to the actual one (by the law of large numbers), the KF variants are restricted to approximate only the first two statistical moments of this PDF. Thus, the KF-based approach is prone to exhibit inferior performance depending on the deviation from its underlying assumptions.

Unfortunately, the high dimensionality of the problem at hand renders the sequential Monte Carlo methods and the UKF computationally inefficient [19]. For that reason, the estimation scheme adopted herein relies on the IEKF [21] discussed in the Introduction. This variant of the EKF is known to better handle nonlinearities. In particular, the IEKF is shown to outperform the UKF in such cases where the state or at least the part of it that causes the nonlinearity in the measurement model is instantaneously fully observable [22]. As the EKF and its variants are widely used for nonlinear estimation (see, [21]), we omit the complete algorithmic details of the IEKF; we nevertheless point out that the Jacobians associated with the process and observation models are computed numerically at each filtering iteration using standard schemes for approximating derivatives. In our derivations, we denote by $\hat{\mathbf{x}}_{k|k}$ and $\hat{\mathbf{x}}_{k|k-1}$ the IEKF-computed mean at time k based on either k or $k-1$ observations, respectively.

A. Coping With Inertia Uncertainties

A major consideration in this paper is the modeling uncertainty associated with the unknown target spacecraft inertia tensor. It was previously shown that such an uncertainty considerably affects the numerical stability of angular-rate estimation schemes owing to the sensitivity of the Euler equations [23]–[25]. In this paper, we solve this issue by using an external estimator interlaced with the main filtering algorithm. In particular, we develop an inertia identification scheme that involves the evaluation of a likelihood score over a finite set of possibilities $\mathcal{I} = \{I_i\}_{i=1}^{N_I}$. As described in the following, this procedure relies on the innovation process of the IEKF. The resulting scheme consists of several IEKFs working in parallel, each is aimed at approximating

$$\hat{\mathbf{x}}_{k|k}(I_i) \approx E\{\mathbf{x}_k | \mathcal{Z}_{1:k}, I_i\} \quad (20)$$

where $\hat{\mathbf{x}}_{k|k}(I_i)$ denotes the IEKF-computed mean assuming a hypothetical inertia tensor I_i .

B. Robust Relative State Filtering

Our filtering scheme consists of several IEKFs working simultaneously. Each filter employs the previously described observation and process models and assumes a hypothetical target inertia tensor I_i . The statistics obtained from the filters are then used for computing a MAP inertia estimator. The detailed estimation procedure is derived next.

In the following, we assume that the inertia tensor, I_i , is a random parameter with some known prior PDF $p(I = I_i)$. Recalling Bayes' rule, we may write

$$p(I_i | \mathcal{Z}_{1:T}) \propto p(\mathcal{Z}_{1:T} | I_i) p(I = I_i) \quad (21)$$

where $p(\mathcal{Z}_{1:T} | I_i)$ is the likelihood of the observation history conditioned on the inertia tensor. It is readily recognized that the likelihood can be written using the factorization

$$p(\mathcal{Z}_{1:T} | I_i) = p(\mathcal{Z}_1 | I_i) \prod_{k=2}^T p(\mathcal{Z}_k | I_i, \mathcal{Z}_{1:k-1}). \quad (22)$$

Expanding the right-hand side of (22) yields

$$\begin{aligned} p(\mathcal{Z}_k | I_i, \mathcal{Z}_{1:k-1}) &= \int \int p(\mathcal{Z}_k | \mathbf{x}_k) p(\mathbf{x}_k | \mathbf{x}_{k-1}, I_i) \\ &\quad \times p(\mathbf{x}_{k-1} | \mathcal{Z}_{1:k-1}) d\mathbf{x}_k d\mathbf{x}_{k-1} \\ &= \int p(\mathcal{Z}_k | \mathbf{x}_k) p(\mathbf{x}_k | \mathcal{Z}_{1:k-1}, I_i) d\mathbf{x}_k \\ &= E \{ p(\mathcal{Z}_k | \mathbf{x}_k) | \mathcal{Z}_{1:k-1}, I_i \} \end{aligned} \quad (23)$$

where the likelihood $p(\mathcal{Z}_k | \mathbf{x}_k)$ and the transition PDF $p(\mathbf{x}_k | \mathbf{x}_{k-1}, I_i)$ are exclusively specified by the observation and process models (18) and (4), respectively. The expectation on the right hand side of (23) is performed with respect to the propagated PDF $p(\mathbf{x}_k | \mathcal{Z}_{1:k-1}, I_i)$. In general, a straightforward computation of this expectation is impractical even if the propagated PDF is known, as it essentially involves a nonlinear function of the state, $p(\mathcal{Z}_k | \mathbf{x}_k)$. This difficulty can be alleviated by retaining the first few terms in a Taylor expansion of the likelihood $p(\mathcal{Z}_k | \mathbf{x}_k)$. Hence

$$\begin{aligned} \hat{p}(\mathcal{Z}_k | \mathbf{x}_k) &= p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)) + (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}(I_i))^T \\ &\quad \times \left[\frac{\partial p(\mathcal{Z}_k | \mathbf{x}_k)}{\partial \mathbf{x}_k} \right]_{\hat{\mathbf{x}}_{k|k-1}(I_i)} + \frac{1}{2} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}(I_i))^T \\ &\quad \times \left[\frac{\partial^2 p(\mathcal{Z}_k | \mathbf{x}_k)}{\partial \mathbf{x}_k \partial \mathbf{x}_k} \right]_{\hat{\mathbf{x}}_{k|k-1}(I_i)} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}(I_i)) \end{aligned} \quad (24)$$

where the expansion reference point is assumed to be the i th IEKF propagated mean at time $k-1$, $\hat{\mathbf{x}}_{k|k-1}(I_i)$. The notations $[\partial p(\mathcal{Z}_k | \mathbf{x}_k) / \partial \mathbf{x}_k]_{\hat{\mathbf{x}}_{k|k-1}(I_i)}$ and $[\partial^2 p(\mathcal{Z}_k | \mathbf{x}_k) / \partial \mathbf{x}_k \partial \mathbf{x}_k]_{\hat{\mathbf{x}}_{k|k-1}(I_i)}$ stand for the Jacobian and Hessian of $p(\mathcal{Z}_k | \mathbf{x}_k)$ with respect to $\mathbf{x}_k$ computed at $\hat{\mathbf{x}}_{k|k-1}(I_i)$, respectively. Now, assuming $\hat{\mathbf{x}}_{k|k-1}(I_i) \approx E \{ \mathbf{x}_k | \mathcal{Z}_{1:k-1}, I_i \}$ and substituting the approximated likelihood (24) into (23) yields (25), shown at the bottom of the page, where $\text{Tr}[\cdot]$ denotes the matrix trace. In addition, if $\text{Tr}[\text{cov}(\mathbf{x}_k | \mathcal{Z}_{1:k-1}, I_i)]$ is sufficiently small, then

$$p(\mathcal{Z}_k | I_i, \mathcal{Z}_{1:k-1}) \approx p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)). \quad (26)$$

It is seen that the likelihood of interest (22) can be approximated using a factorization of the right hand side in (26) over

the time interval $[1, T]$, that is

$$\begin{aligned} p(\mathcal{Z}_{1:T} | I_i) &= p(\mathcal{Z}_1 | I_i) \prod_{k=2}^T p(\mathcal{Z}_k | I_i, \mathcal{Z}_{1:k-1}) \\ &\approx \prod_{k=1}^T p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)). \end{aligned} \quad (27)$$

Let

$$\hat{p}(\mathcal{Z}_{1:T} | I_i) := \prod_{k=1}^T p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)). \quad (28)$$

Then, substituting (18) into (28) while assuming the observation noise $\mathbf{n}_j$ is Gaussian, yields

$$\begin{aligned} \log \hat{p}(\mathcal{Z}_{1:T} | I_i) &= \sum_{k=1}^T \log p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)) \\ &= \sum_{k=1}^T \sum_{j=1}^N \log p_{\mathbf{n}_j}(\mathbf{z}_{jk} - h_j(\hat{\mathbf{x}}_{k|k-1}(I_i))) \\ &= T \log c - \frac{1}{2} \sum_{k=1}^T \sum_{j=1}^N (\mathbf{z}_{jk} - h_j(\hat{\mathbf{x}}_{k|k-1}(I_i)))^T \\ &\quad \times R_k^{-1} (\mathbf{z}_{jk} - h_j(\hat{\mathbf{x}}_{k|k-1}(I_i))) \end{aligned} \quad (29)$$

which essentially consists of computing the innovations of each filter over the time interval $[1, T]$. The most probable inertia hypothesis is then considered as the MAP estimate

$$\hat{I}_{\text{MAP}} = \arg \max_{I_i \in \mathcal{I}} \log \hat{p}(\mathcal{Z}_{1:T} | I_i) + \log p(I = I_i). \quad (30)$$

In practice, (30) is used to identify the most consistent IEKF (i.e., having the lowest innovation variance) from which the relative state estimates is to be considered. For completeness, the robust multiple IEKFs algorithm is summarized next.

C. Boundedness Analysis: Deviations from Optimality

As the proposed robust filtering scheme relies on an approximate likelihood PDF (26), its statistical consistency (i.e., convergence toward the sought-after parameters as the amount of observations increases [26]) is questionable. The following analysis is aimed at elucidating the statistical behavior of the inertia estimation scheme in this regard:

$$\begin{aligned} p(\mathcal{Z}_k | I_i, \mathcal{Z}_{1:k-1}) &\approx E \{ \hat{p}(\mathcal{Z}_k | \mathbf{x}_k) | \mathcal{Z}_{1:k-1}, I_i \} \\ &= p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)) + E \left\{ (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}(I_i))^T \left[\frac{\partial p(\mathcal{Z}_k | \mathbf{x}_k)}{\partial \mathbf{x}_k} \right]_{\hat{\mathbf{x}}_{k|k-1}(I_i)} | \mathcal{Z}_{1:k-1}, I_i \right\} \\ &\quad + \frac{1}{2} E \left\{ \text{Tr} \left[\left[\frac{\partial^2 p(\mathcal{Z}_k | \mathbf{x}_k)}{\partial \mathbf{x}_k \partial \mathbf{x}_k} \right]_{\hat{\mathbf{x}}_{k|k-1}(I_i)} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}(I_i)) (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}(I_i))^T \right] | \mathcal{Z}_{1:k-1}, I_i \right\} \\ &= p(\mathcal{Z}_k | \hat{\mathbf{x}}_{k|k-1}(I_i)) + \frac{1}{2} \text{Tr} \left\{ \left[\frac{\partial^2 p(\mathcal{Z}_k | \mathbf{x}_k)}{\partial \mathbf{x}_k \partial \mathbf{x}_k} \right]_{\hat{\mathbf{x}}_{k|k-1}(I_i)} \text{cov}(\mathbf{x}_k | \mathcal{Z}_{1:k-1}, I_i) \right\} \end{aligned} \quad (25)$$

Algorithm 1 A Single Cycle of the Robust Multiple IEKFs

-
- 1: Initialization: $\hat{\mathbf{x}}_0(I_i) = \mathbf{0}$, $I_i \sim p(I)$, $i = 1, \dots, N_I$
 - 2: At time k perform the following
 - 3: **for** $i = 1, \dots, N_I$ **do**
 - 4: Perform a single IEKF iteration using an hypothetical inertia I_i to obtain $\hat{\mathbf{x}}_{k|k-1}(I_i)$ and $\hat{\mathbf{x}}_{k|k}(I_i)$
 - 5: Compute the posterior score $\log \hat{p}(\mathcal{Z}_{k-T:k} \mid I_i)$ using (29)
 - 6: **end for**
 - 7: Compute the MAP estimate using (30) and set

$$i^* = \{i \mid I_i = \hat{I}_{\text{MAP}}, i = 1, \dots, N_I\}$$
 - 8: Set the algorithm's output as $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k}(I_{i^*})$
-

1) *Preliminaries:* Before proceeding, we briefly review some of the notions that are used in the subsequent analysis. The following definitions are based on the ones presented in [27], [28].

Definition 1 (Absolute Continuity): A probability distribution $P(x)$ is absolutely continuous with respect to the probability distribution $Q(x)$ if $P(x \in A) = 0$ for every set A for which $Q(x \in A) = 0$ (i.e., the support of $P(x)$ includes the support of $Q(x)$).

If, in addition, the underlying probability distributions are assigned only finite values, then the definition of absolute continuity above implies that the Radon–Nykodim derivative [27] exists and is finite. This derivative is defined as

$$\frac{dP(x)}{dQ(x)} = \frac{p(x)}{q(x)} \quad (31)$$

where $p(x)$ and $q(x)$ are the corresponding probability densities. In other words, absolute continuity of $P(x)$ with respect to $Q(x)$ guarantees that (31) exists.

Definition 2 (Comparability): Two probability distributions $P(x)$ and $Q(x)$ are said to be comparable if and only if $P(x)$ is absolutely continuous with respect to $Q(x)$, and $Q(x)$ is absolutely continuous with respect to $P(x)$.

Definition 3 (Measurable Random Variable): Let x and y be two random variables. The random variable x is y -measurable if its realization is completely determined by the outcome of y .

Definition 4 (Jensen's Inequality): Let x be a random variable, and let also $\phi(x)$ be a convex function. Then

$$E\{\phi(x)\} \geq \phi(E\{x\}). \quad (32)$$

Definition 5 (Kullback-Leibler (KL) Divergence): The KL divergence quantifies the discrepancy between two comparable probability distributions $P(x)$ and $Q(x)$. Its continuous version, which is defined using the corresponding probability densities, is given by

$$\text{KL}(p\|q) = \int_x p(x) \frac{\log p(x)}{\log q(x)} dx = E\left\{\frac{\log p(x)}{\log q(x)}\right\} \quad (33)$$

where the expectation is performed with respect to $p(x)$. This measure satisfies $\text{KL}(p\|q) \geq 0$, and can be shown to violate the triangle inequality (and therefore is not considered as a metric).

Definition 6 (Super Martingale): A discrete-time super Martingale $\mathcal{M}_k$ with respect to a random sequence $\mathcal{Z}_{1:k}$ is a random process satisfying

$$E\{\mathcal{M}_k \mid \mathcal{Z}_{1:k-1}\} \leq \mathcal{M}_{k-1}.$$

Theorem 1 (Super Martingale Convergence): A super Martingale $\mathcal{M}_k$ having integrable sequences, in the sense $\sup_k E\{|\mathcal{M}_k|^d\} < \infty$ for some $d > 1$, converges almost surely (i.e., with probability 1) to a random variable $\mathcal{M}_\infty$, satisfying $E\{\mathcal{M}_\infty\} < \infty$.

2) *Main Results:* The main results in this section are provided below.

Theorem 2: Let $\hat{I}_k$ be a $\mathcal{Z}_{1:k}$ -measurable estimator of I (e.g., the MMSE estimator $E\{I \mid \mathcal{Z}_{1:k}\}$ or the MAP estimator) represented in a vector form (i.e., concatenation of its columns within a vector). Let also $E\{\hat{I}_k\}$ and $\hat{E}\{\hat{I}_k\}$ be the expected values of $\hat{I}_k$ when this estimator is computed using the optimal distribution, $P(\mathcal{Z}_{1:k}, I)$, and an approximate one, $\hat{P}(\mathcal{Z}_{1:k}, I)$, respectively. If, in addition, these two probability distributions are comparable, or the approximate distribution is absolutely continuous with respect to the optimal one, then the following holds:

$$\left\|E\{\hat{I}_k\} - \hat{E}\{\hat{I}_k\}\right\|_2^2 \leq M_k \text{Tr}[\text{cov}(\delta I_k)] \quad (34)$$

where $\delta I_k = I - \hat{I}_k$ is the inertia estimation error and M_k is some positive ($I, \mathcal{Z}_{1:k}$)-measurable random variable that is almost surely finite.

Proof: Let I^* be some deterministic inertia vector (i.e., concatenation of its columns within a vector). Using the smoothing property of the conditional expectation we notice that

$$\begin{aligned} & \left\|E\{I\} - \hat{E}\{\hat{I}\}\right\|_2^2 \\ &= \left\|E\left\{E\{\hat{I} \mid I\} - \hat{E}\{\hat{I} \mid I\}\right\}\right\|_2^2 \\ &= \left\|E\left\{E\{I - \hat{I} - I^* \mid I\} - \hat{E}\{I - \hat{I} - I^* \mid I\}\right\}\right\|_2^2 \\ &= \left\|E\left\{\int (I - \hat{I} - I^*) p(\mathcal{Z}_{1:k} \mid I) d\mathcal{Z}_{1:k} \right. \right. \\ &\quad \left. \left. - \int (I - \hat{I} - I^*) \hat{p}(\mathcal{Z}_{1:k} \mid I) d\mathcal{Z}_{1:k} \right\}\right\|_2^2 \\ &= \left\|E\left\{\int (I - \hat{I} - I^*) \left(1 - \frac{\hat{p}(\mathcal{Z}_{1:k} \mid I)}{p(\mathcal{Z}_{1:k} \mid I)}\right) \right. \right. \\ &\quad \left. \left. \times p(\mathcal{Z}_{1:k} \mid I) d\mathcal{Z}_{1:k} \right\}\right\|_2^2 \\ &= \left\|E\left\{E\left\{\left(I - \hat{I} - I^*\right) \left(1 - \frac{\hat{p}(\mathcal{Z}_{1:k} \mid I)}{p(\mathcal{Z}_{1:k} \mid I)}\right) \mid I\right\}\right\}\right\|_2^2. \end{aligned} \quad (35)$$

Further invoking Jensen's inequality yields

$$\begin{aligned} & \left\|E\left\{E\left\{\left(I - \hat{I} - I^*\right) \left(1 - \frac{\hat{p}(\mathcal{Z}_{1:k} \mid I)}{p(\mathcal{Z}_{1:k} \mid I)}\right) \mid I\right\}\right\}\right\|_2^2 \\ & \leq E\left\{E\left\{\left\|I - \hat{I} - I^*\right\|_2^2 \left(1 - \frac{\hat{p}(\mathcal{Z}_{1:k} \mid I)}{p(\mathcal{Z}_{1:k} \mid I)}\right)^2 \mid I\right\}\right\} \end{aligned}$$

$$\leq E \left\{ \left\| I - \hat{I} - I^* \right\|_2^2 \right\} M_k \quad (36)$$

where

$$M_k \triangleq \left[\sup_{\mathcal{Z}_{1:k}, I} \left| 1 - \frac{\hat{p}(\mathcal{Z}_{1:k} | I)}{p(\mathcal{Z}_{1:k} | I)} \right| \right]^2. \quad (37)$$

Setting $I^* = E \{ I - \hat{I} \}$ in (36), both (35) and (36) yield

$$\left\| E \{ I \} - \hat{E} \{ \hat{I} \} \right\|_2^2 \leq E \left\{ \left\| \delta I - E \{ \delta I \} \right\|_2^2 \right\} M_k \quad (38)$$

where $\delta I = I - \hat{I}$. Finally, (38) implies

$$\begin{aligned} & \left\| E \{ I \} - \hat{E} \{ \hat{I} \} \right\|_2^2 \\ & \leq E \left\{ \text{Tr} \left[(\delta I - E \{ \delta I \}) (\delta I - E \{ \delta I \})^T \right] \right\} M_k \\ & = \text{Tr} \left[E \left\{ (\delta I - E \{ \delta I \}) (\delta I - E \{ \delta I \})^T \right\} \right] M_k \\ & = \text{Tr} [\text{cov}(\delta I)] M_k \end{aligned} \quad (39)$$

which thereby completes the proof. $\blacksquare$

Theorem 3: For sufficiently large k , the following is satisfied:

$$\sqrt{M_k} \rightarrow \left| 1 - \left(\exp \left\{ -KL(p(\mathcal{Z} | I) \parallel \hat{p}(\mathcal{Z} | I)) + \mathcal{O}(1/\sqrt{k}) \right\} \right)^k \right| \text{ almost surely.} \quad (40)$$

where $KL(p \parallel \hat{p}) > 0$ denotes the Kullback-Leibler divergence between the two comparable, yet distinct, PDFs $p(\cdot)$ and $\hat{p}(\cdot)$. In particular

$$\sqrt{M_k} \xrightarrow{k \rightarrow \infty} 1 \text{ almost surely} \quad (41)$$

Proof: We begin by showing that the ratio

$$\mathcal{M}_k(I) = \frac{\hat{p}(\mathcal{Z}_{1:k} | I)}{p(\mathcal{Z}_{1:k} | I)} \quad (42)$$

is a positive super Martingale with respect to $\mathcal{Z}_{1:k-1}$. In detail

$$\begin{aligned} & E \{ \mathcal{M}_k(I) | \mathcal{Z}_{1:k-1}, I \} \\ & = E \left\{ \frac{\hat{p}(\mathcal{Z}_{1:k} | I)}{p(\mathcal{Z}_{1:k} | I)} | \mathcal{Z}_{1:k-1}, I \right\} \\ & = E \left\{ \frac{\hat{p}(\mathcal{Z}_k | I)}{p(\mathcal{Z}_k | I)} \mathcal{M}_{k-1}(I) | \mathcal{Z}_{1:k-1}, I \right\} \\ & = E \left\{ \frac{\hat{p}(\mathcal{Z}_k | I)}{p(\mathcal{Z}_k | I)} \right\} \mathcal{M}_{k-1}(I) = \mathcal{M}_{k-1}(I). \end{aligned} \quad (43)$$

As $\mathcal{M}_k$ is almost surely finite (from the comparability of the two likelihoods), it converges to a random variable $\mathcal{M}_\infty$ where $E \{ \mathcal{M}_\infty \} < \infty$. It is fairly easy to derive an expression of $\mathcal{M}_k$ for large enough, possibly infinite, k . The elementary argument proceeds as follows. Recalling the strong law of large numbers we recognize that

$$\begin{aligned} & k^{-1} \log \mathcal{M}_k(I) \\ & = k^{-1} \log \frac{\hat{p}(\mathcal{Z}_{1:k} | I)}{p(\mathcal{Z}_{1:k} | I)} \\ & = k^{-1} \log \prod_{i=1}^k \frac{\hat{p}(\mathcal{Z}_i | I)}{p(\mathcal{Z}_i | I)} \end{aligned}$$

$$\begin{aligned} & = k^{-1} \sum_{i=1}^k \log \frac{\hat{p}(\mathcal{Z}_i | I)}{p(\mathcal{Z}_i | I)} \xrightarrow{k \rightarrow \infty} E \left\{ \frac{\hat{p}(\mathcal{Z} | I)}{p(\mathcal{Z} | I)} | I \right\} \\ & = - \int p(\mathcal{Z} | I) \log \frac{p(\mathcal{Z} | I)}{\hat{p}(\mathcal{Z} | I)} d\mathcal{Z} \\ & = -\text{KL}(p(\mathcal{Z} | I) \parallel \hat{p}(\mathcal{Z} | I)) \text{ almost surely} \end{aligned} \quad (44)$$

which is merely an empirical estimate of the relative entropy between the actual and approximated likelihoods, $p(\mathcal{Z} | I)$ and $\hat{p}(\mathcal{Z} | I)$. The central limit theorem (CLT) [29] is an aid for evaluating the accuracy of such an estimator. In particular, the CLT asserts

$$k^{-1} \log \mathcal{M}_k(I) = -\text{KL}(p(\mathcal{Z} | I) \parallel \hat{p}(\mathcal{Z} | I)) + \mathcal{O}(1/\sqrt{k}) \quad (45)$$

for sufficiently large k , where the order symbol stands for some zero-mean Gaussian random variable with a standard deviation $\sigma/\sqrt{k}$. Finally, the theorem readily follows from both (37) and (45) (which essentially yields $\mathcal{M}_\infty = 0$). $\blacksquare$

The striking observation conveyed by both these theorems is that the effect of any relaxation whatsoever diminishes with the amount of observations. To be more precise, deviation from optimality of any approximate scheme satisfying the regularity conditions of Theorem 2 is bounded by the optimal inertia estimation error covariance. This, in turn, gives rise to the following premise. If the optimal estimator is statistically consistent, then any approximate scheme satisfying the conditions of Theorem 2 is likewise statistically consistent. The rate of convergence toward the optimal solution is dictated by (40), in which the discrepancy between the relaxed and the exact likelihoods (quantified by the Kullback–Leibler divergence) plays a major role.

IV. ESTIMABILITY UNDER MODELING UNCERTAINTIES

By defining $\Delta \rho_i \triangleq \rho_i - \rho_0$, Eq. (13) may be re-written into

$$\Delta \rho_{ik} = D(\mathbf{q}_k) \mathbf{P}_i + \mathbf{v}_{ik} \quad (46)$$

where a noise sequence, $\mathbf{v}_{ik}$, is introduced to account for modeling uncertainties. Equation (46) bares a clear resemblance to the vector observations model, commonly used for attitude estimation [30], [31]. This model relates a reference vector, $\mathbf{P}_i$, to its noisy body-frame representation at time k , $\Delta \rho_{ik}$. A key assumption underlying the vector observations model is that the reference vector and associated reference frame are accurately known, as otherwise the attitude turns out to be unobservable [32]. However, in the problem at hand the reference vectors in (46) are not accurately known, because the target's CM, to which the target's body frame is anchored, is unknown.

Therefore, to investigate the estimability of the relative attitude, and possibly the feature points constituting the target's structure, we resort to an information-based approach. In principle, we use the notion of (Fisher) information submatrix [33] to prove two fundamental conjectures. The first one, which is summarized by the subsequent proposition, quantifies the reduction in the information associated with the attitude channel because of the uncertain reference frame. This degenerative effect is shown to diminish with the amount of observations, as elucidated by a successive corollary.

$$B_k^T = E \left\{ \begin{array}{c} \partial (D(\mathbf{q}_1)^T \Delta \rho_{11})^T / \partial \mathbf{q}_1, \dots, \partial (D(\mathbf{q}_1)^T \Delta \rho_{1N})^T / \partial \mathbf{q}_1 \\ \vdots \\ \partial (D(\mathbf{q}_k)^T \Delta \rho_{k1})^T / \partial \mathbf{q}_k, \dots, \partial (D(\mathbf{q}_k)^T \Delta \rho_{kN})^T / \partial \mathbf{q}_k \end{array} \right\} \in \mathbb{R}^{4k \times 3N}. \quad (48)$$

$$J = \left[\begin{array}{c|c} A_k & B_k \\ \hline B_k^T & C_k \end{array} \right] \triangleq \left[\begin{array}{c|c} E \{ -\partial^2 \log \bar{p}_k / \partial \mathbf{P} \partial \mathbf{P} \} & E \{ -\partial^2 \log \bar{p}_k / \partial \mathbf{P} \partial \mathbf{q}_{0:k} \} \\ \hline E \{ -\partial^2 \log \bar{p}_k / \partial \mathbf{q}_{0:k} \partial \mathbf{P} \} & E \{ -\partial^2 \log \bar{p}_k / \partial \mathbf{q}_{0:k} \partial \mathbf{q}_{0:k} \} \end{array} \right] \quad (50)$$

Proposition 1: Let σ and σ_p be the standard deviations associated with the observation noise and the feature point locations, respectively. Then the information submatrix associated with the quaternion trajectory, $\mathbf{q}_{0:k} \triangleq [\mathbf{q}_0^T, \dots, \mathbf{q}_k^T]^T$, is given by

$$J_{0:k}^{\mathbf{q}} = \mathcal{J}_{0:k}^{\mathbf{q}} - \frac{\sigma_p^2 \sigma^2}{k\sigma_p^2 + \sigma^2} B_k^T B_k \quad (47)$$

where $\mathcal{J}_{0:k}^{\mathbf{q}}$ is the Fisher information matrix associated with $\mathbf{q}_{0:k}$ had the feature point locations been perfectly known, and see (48), shown at the top of the page.

Proof: We begin by formulating the joint PDF of the quaternion trajectory, $\mathbf{q}_{0:k}$, the concatenation of feature point vectors, $\mathbf{P} \triangleq [\mathbf{P}_1^T, \dots, \mathbf{P}_N^T]^T$, and the observations, by means of the underlying transition and likelihood PDFs. Thus

$$\begin{aligned} \bar{p}_k &\triangleq p(\mathbf{q}_{0:k}, \mathbf{P}, \Delta \rho_{1:k}) \\ &= p(\mathbf{q}_0) p(\mathbf{P}) \prod_{j=1}^k p(\mathbf{q}_j | \mathbf{q}_{j-1}) p(\Delta \rho_j | \mathbf{q}_j, \mathbf{P}) \end{aligned} \quad (49)$$

where $\Delta \rho_{1:k} \triangleq \{\Delta \rho_1, \dots, \Delta \rho_k\}$, and $\Delta \rho_j \triangleq [\Delta \rho_{j1}^T, \dots, \Delta \rho_{jN}^T]^T$. The Fisher information matrix associated with the joint posterior $\bar{p}_k$ can be decomposed as [33] shown in (50) at the top of the page. The information submatrix associated with $\mathbf{q}_{0:k}$ can now be obtained as [33]

$$J_{0:k}^{\mathbf{q}} = C_k - B_k^T A_k^{-1} B_k \quad (51)$$

assuming the inverse exists. Following the rationale in [33], the inverse of this information submatrix can be used for computing the Cramér-Rao lower bound (CRLB) of any unbiased estimator of $\mathbf{q}_{0:k}$. Now, assuming a zero-mean Gaussian observation noise with standard deviation $\sigma \mathbb{I}_{3 \times 3}$ in (46) (which constitutes the likelihood $p(\Delta \rho_k | \mathbf{q}_k, \mathbf{P})$), and a Gaussian distribution of the feature point locations $p(\mathbf{P}) = \mathcal{N}(\mathbf{P} | \boldsymbol{\mu}_p, \sigma_p^2 \mathbb{I}_{3N \times 3N})$, yields (after some simple algebraic manipulations)

$$A_k^{-1} = \frac{\sigma_p^2 \sigma^2}{k\sigma_p^2 + \sigma^2} \mathbb{I}_{3N \times 3N} \quad (52)$$

and the expression for B_k in (48). Finally, recognizing that C_k is merely the Fisher information matrix associated with $\mathbf{q}_{0:k}$, assuming perfectly known feature point locations $\mathbf{P}$, both (51) and (52) yields (47). ■

The next corollary follows immediately from (47).

Corollary 1: There is a critical uncertainty level associated with the feature points (σ_p in (47)) from which both the attitude and the feature points are prone to be nonestimable in the sense that the (Fisher) information content of both these

channels is likely to decrease over some finite time interval. The amount of observations, k , needed for diminishing this undesirable effect increases with the uncertainty level. Nevertheless, the asymptotic information content in the attitude channel approaches the attainable limit had the feature points been perfectly known

$$J_{0:k}^{\mathbf{q}} \xrightarrow{k \rightarrow \infty} \mathcal{J}_{0:\infty}^{\mathbf{q}}. \quad (53)$$

The critical uncertainty level mentioned in Corollary 1 refers to some value of σ_p from which the information content (47) is likely to decrease. Such a behavior is expected for small values of k , in particular when σ_p is larger than the initial uncertainty associated with the attitude (contained within $\mathcal{J}_{0:k}^{\mathbf{q}}$). Nevertheless, as stated in Corollary 1, this effect diminishes with the amount of observations, k .

As the Fisher information submatrix (47) constitutes the posterior CRLB on the estimation error of any unbiased attitude estimator [33], we would expect this bound to follow the aforementioned behavior, however, in a converse manner. That is, we would expect the CRLB to increase for small k 's until some point from which it would consistently decrease. As the uncertainty level σ_p goes up, we would expect an expeditious initial increase.

V. NUMERICAL STUDY

Here, we examine the performance of the filtering algorithm developed in previous sections. We begin by a numerical computation of the CRLB.

A. CRLB Computation

The matrix trace of the CRLB is shown in Fig. 2. The lower bounds on both the attitude and the feature point channels are numerically approximated for the problem at hand using the techniques described in [33] and [14]. Fig. 2 clearly shows that the CRLB indeed behaves as predicted by the qualitative results in the previous section.

B. Monte-Carlo Simulation

An extensive Monte-Carlo (MC) simulation study is conducted for assessing the performance of the derived filtering scheme. The simulations are done using MATLAB. We used functions from the MATLABstatistics toolbox for the generation of the random signals. The simulation flowchart is shown in Fig. 3.

The following example consists of two satellites in low earth orbits. The chaser satellite's eccentricity and semimajor axis

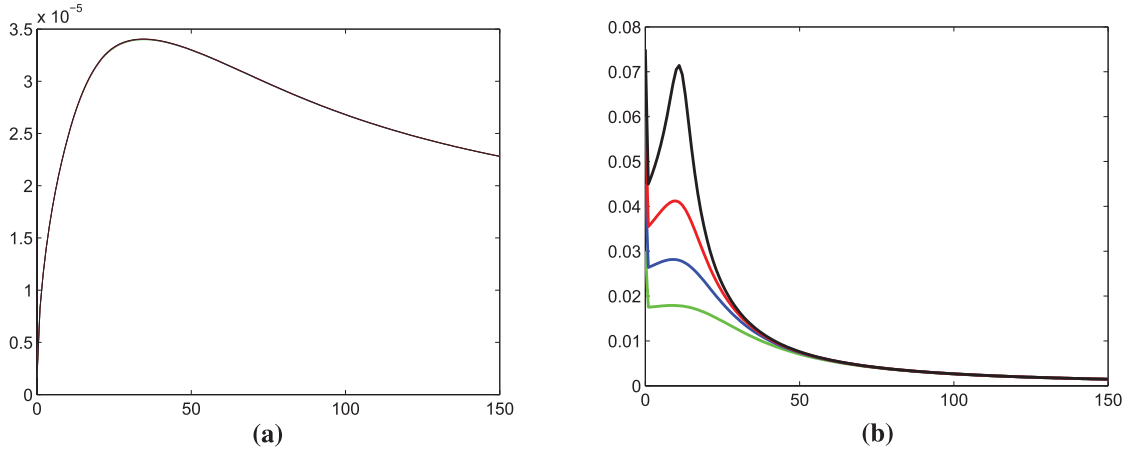


Fig. 2. (a) Quaternion CRLB and (b) feature points CRLB for various feature point uncertainty levels (shown bottom to top): $\sigma_p = 1, 1.5, 2$, and 2.5 cm.

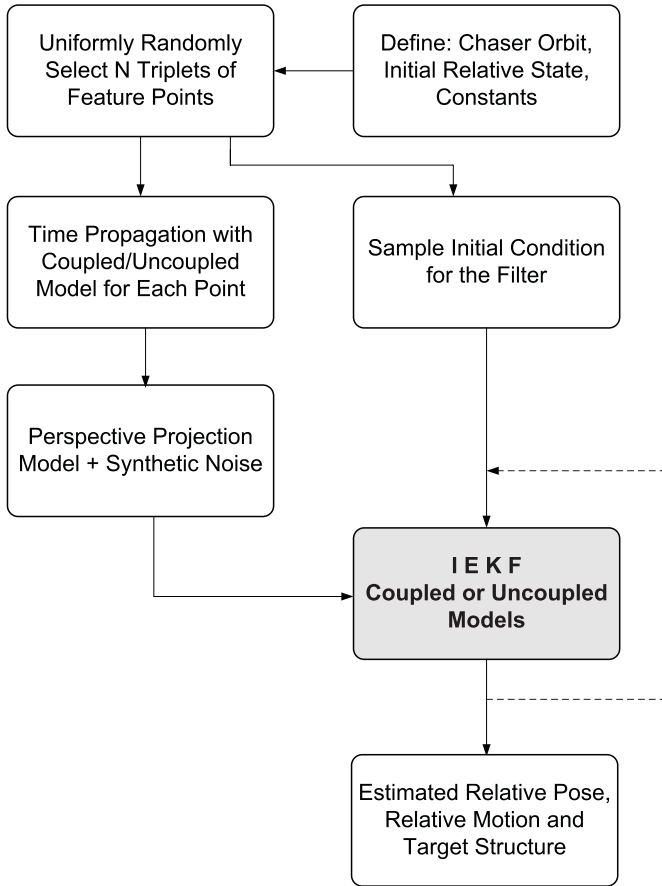


Fig. 3. Simulation flowchart.

are set as $e_C = 0.05$ and $a_C = 7170$ km, respectively. In all runs, the chaser's inertia tensor is assumed to be diagonal, $I_C = \text{diag}(100, 120, 130)$ kg m². The target's inertia tensor, on the other hand, is set as

$$I_T = \begin{bmatrix} 96 & 43 & 26 \\ 43 & 104 & 10 \\ 26 & 10 & 133 \end{bmatrix} \text{ kg m}^2. \quad (54)$$

The chaser satellite is equipped with a stereovision system consisting of two cameras with a baseline $b = 1$ m. It is

assumed that there are exactly ten feature points, the locations of which are uniformly spread over the target

$$P_i^{(j)} \sim \mathcal{U}(-1.5 \text{ m}, 1.5 \text{ m}), \quad i = 1, \dots, 10; \quad j = 1, 2, 3. \quad (55)$$

The two modeling approaches, namely the coupled and uncoupled models, are integrated using the MATLAB ode45 function to generate the translational trajectories of the target feature points and the relative rotational motion. In the uncoupled model, the relative initial conditions are set as follows:

$$\rho_0(0) = [10, 100, 10]^T \text{ m} \quad (56a)$$

$$\dot{\rho}_0(0) = [0.01, -0.0225, -0.01]^T \text{ m/s} \quad (56b)$$

$$\omega_C(0) = [0, 0, 0.0656]^T \text{ deg/s} \quad (56c)$$

$$\omega_T(0) = [0.1, 0.1, 0.1]^T \text{ deg/s}. \quad (56d)$$

The initial relative attitude is set as $\mathbf{q} = [0, 0, 0, 1]^T$ in all runs. In the coupled model, the initial conditions for the trajectories of each one of the feature points are computed using the relations

$$\begin{aligned} \rho_i(0) &= \rho_0(0) + D(\mathbf{q})\mathbf{P}_i \\ \dot{\rho}_i(0) &= \dot{\rho}_0(0) + \omega(0) \times \mathbf{P}_i, \quad i = 1, \dots, 10. \end{aligned} \quad (57)$$

In all cases, the filter initial conditions are drawn from a Gaussian distribution, that is, $\hat{\mathbf{x}}_0 \sim \mathcal{N}(\hat{\mathbf{x}}_0, W(0))$ using the MATLAB function `mvnrnd`, where the initial covariance is chosen to be

$$W(0) = \text{diag} \left(\left[\sigma_{W\rho_1}^2, \dots, \sigma_{W\rho_N}^2, \sigma_{W\dot{\rho}_1}^2, \dots, \sigma_{W\dot{\rho}_N}^2, \sigma_{W\omega}^2, \sigma_{W\mathbf{q}}^2, \sigma_{W\mathbf{P}_1}^2, \dots, \sigma_{W\mathbf{P}_N}^2 \right] \right) \quad (58)$$

with $\sigma_{W\rho}^2 = [1, 1, 1] \text{ m}^2$, $\sigma_{W\dot{\rho}}^2 = [1, 1, 1] \text{ m}^2/\text{s}^2$, $\sigma_{W\omega}^2 = [1, 1, 1] \text{ deg}^2/\text{s}^2$, $\sigma_{W\mathbf{q}}^2 = [1, 1, 1, 1] \cdot 10^{-5}$ and $\sigma_{W\mathbf{P}_F}^2 = [1, 1, 1] \text{ m}^2$. Noisy observations are modeled by adding noise to the perspective projection of the simulated trajectories of each one of the feature points. The observation noise is assumed to be zero-mean Gaussian with a standard deviation of 10^{-5} radians. The MATLAB function `mvnrnd` is used for generating the synthetic noise. The robust multiple IEKF scheme uses five hypotheses of the target inertia tensor. These

hypotheses are randomly sampled from a Wishart distribution [34] in the following manner:

$$I_i = I_T + \delta I_i, \quad \delta I_i \sim \mathcal{W}(\sigma_{I_T}, \text{DF}), \quad i = 1, \dots, 5 \quad (59)$$

with $\sigma_{I_T} = (1/3)I_T I_T^T$ and $\text{DF} = 3$.

In the remainder of this section, and in the next section, we denote by $\hat{\xi}$ the estimate of ξ , and use the following definition of estimation errors. Relative position errors are defined as

$$e_{\rho_i} \triangleq \sqrt{(\rho_i^{(1)} - \hat{\rho}_i^{(1)})^2 + (\rho_i^{(2)} - \hat{\rho}_i^{(2)})^2 + (\rho_i^{(3)} - \hat{\rho}_i^{(3)})^2}$$

$$e_{\rho} = \sum_{i=1}^N (e_{\rho_i}) \quad (60)$$

where, in the uncoupled model $i = 0$. Relative velocity errors are defined as

$$e_{\dot{\rho}_i} \triangleq \sqrt{(\dot{\rho}_i^{(1)} - \dot{\hat{\rho}}_i^{(1)})^2 + (\dot{\rho}_i^{(2)} - \dot{\hat{\rho}}_i^{(2)})^2 + (\dot{\rho}_i^{(3)} - \dot{\hat{\rho}}_i^{(3)})^2}$$

$$e_{\dot{\rho}} = \sum_{i=1}^N (e_{\dot{\rho}_i}) \quad (61)$$

where again in the uncoupled model $i = 0$. Denoting $\omega = [\omega_1, \omega_2, \omega_3]^T$, the angular velocity estimation errors are defined as

$$e_{\omega} \triangleq \sqrt{(\omega_1 - \hat{\omega}_1)^2 + (\omega_2 - \hat{\omega}_2)^2 + (\omega_3 - \hat{\omega}_3)^2}. \quad (62)$$

The error quaternion is defined as $\mathbf{q}_e = \mathbf{q} \otimes \hat{\mathbf{q}}^{-1}$ (recall that $\mathbf{q}^{-1} \triangleq \mathbf{q}^* / \|\mathbf{q}\|^2$, $\mathbf{q}^*$ being the conjugate of $\mathbf{q}$). The symbol $\otimes$ denotes the product of two quaternions as described in [30]. The attitude estimation error is then defined as

$$e_{\theta} = 2 \cos^{-1}(q_{e4}) \quad (63)$$

where q_{e4} is the scalar part of $\mathbf{q}_e$. Finally, to define the estimation error of the feature point locations on the target (which quantify the quality of structure recovery), we denote $\mathbf{P}_i = [P_i^{(1)}, P_i^{(2)}, P_i^{(3)}]$ and then

$$e_{P_i} \triangleq \sqrt{(P_i^{(1)} - \hat{P}_i^{(1)})^2 + (P_i^{(2)} - \hat{P}_i^{(2)})^2 + (P_i^{(3)} - \hat{P}_i^{(3)})^2}$$

$$e_P = \sum_{i=1}^N (e_{P_i}). \quad (64)$$

C. Results

The first tests include a comparison between the coupled and uncoupled models based on 100 MC runs. In these runs, a single IEKF is employed for estimating the relative pose, motion and structure assuming that the target's inertia is known. This assumption will be relieved later on.

The initial conditions of the estimator are drawn in each run (in addition to the noise sequences) based on a Gaussian distribution. The standard deviations are as follows: 1) 1 m for each relative position component and each feature point; 2) 1 m/s for each relative velocity component; 3) 1 deg/s for each relative angular velocity component; and 4) 10^5 for each quaternion component.

The results of the comparison are shown in Fig. 4, showing the 50, 70, and 90 percentiles of the estimation errors in the relative position [Fig. 4(a)], relative velocity [Fig. 4(b)], relative attitude [Fig. 4(c)], relative angular velocity [Fig. 4(d)], and feature points [Fig. 4(e)]. Evidently, the estimation scheme utilizing the coupled model has two marked features: It is able to estimate the relative position with an accuracy of a few centimeters (about 2 cm in the median) and the feature points down to a few millimeters (about 2 mm in the median). The uncoupled scheme, on the other hand, provides more accurate estimates of the relative translational and angular velocities, and better estimates of the relative attitude (about 0.25 deg in the median). The coupled model is designed to capture the exact dynamics of each feature point, and hence provides better accuracy in these variables. However, this does not necessarily imply that the other states would be better estimated, because the EKF is only an approximate nonlinear filter. Generally, the coupled model is more computationally extensive than the uncoupled model because of the larger dimension of the state vector (see (11)). The effect of incorporating the robust multiple IEKF estimator is shown in Fig. 5, which shows the distribution of the relative attitude errors of the plain [Fig. 5(a)] and robust [Fig. 5(b)] filtering schemes through the 45, 50, 70, 90, and 95 percentiles. These graphs are generated based on 1000 MC runs, in which the inertia tensor is drawn according to (59). The main difference between the two estimation algorithms is manifested through the 90 and 95 percentile lines. The robust multiple IEKFs algorithm provides 0.75° of attitude estimation accuracy in practically all of the MC runs, whereas the plain IEKF cannot guarantee relative attitude estimation that is better than 1° . These results demonstrate the feasibility of stereovision-based spacecraft relative state estimation under modeling uncertainty.

D. Sensitivity to Measurement Noise

As an important validation of the methodology considered herein, an analysis of the estimation performance with varying measurement noise levels is performed for the uncoupled model. The nonlinearity inherent in the measurement process implies a non-Gaussian measurement noise, and that the nonlinearity might affect the actual space application because of the large relative distances involved. Although the estimation methodology proposed herein assumes a Gaussian noise model, the subsequent numerical results indicate that this discrepancy is likely to have a limited impact on the performance of the proposed scheme (for a discussion on noise models and typical noise levels see [35], [36]; in particular, see [37] for a discussion on the pros and cons of using a Gaussian distribution for measurement noise modelling).

Increased noise levels can result from poor sensors, mechanical vibrations, or may be thought of as a residual effect of the self-attitude estimation errors of the chaser satellite. Thus, in addition to the original measurement noise covariance of 10^{-5} radians, simulations with measurement noise covariances of 5×10^{-5} and 10^{-4} are carried out. Fig. 6 compares the results of 100 MC simulation runs for the various noise levels (90th percentile). It is evident that the relative distance estimation

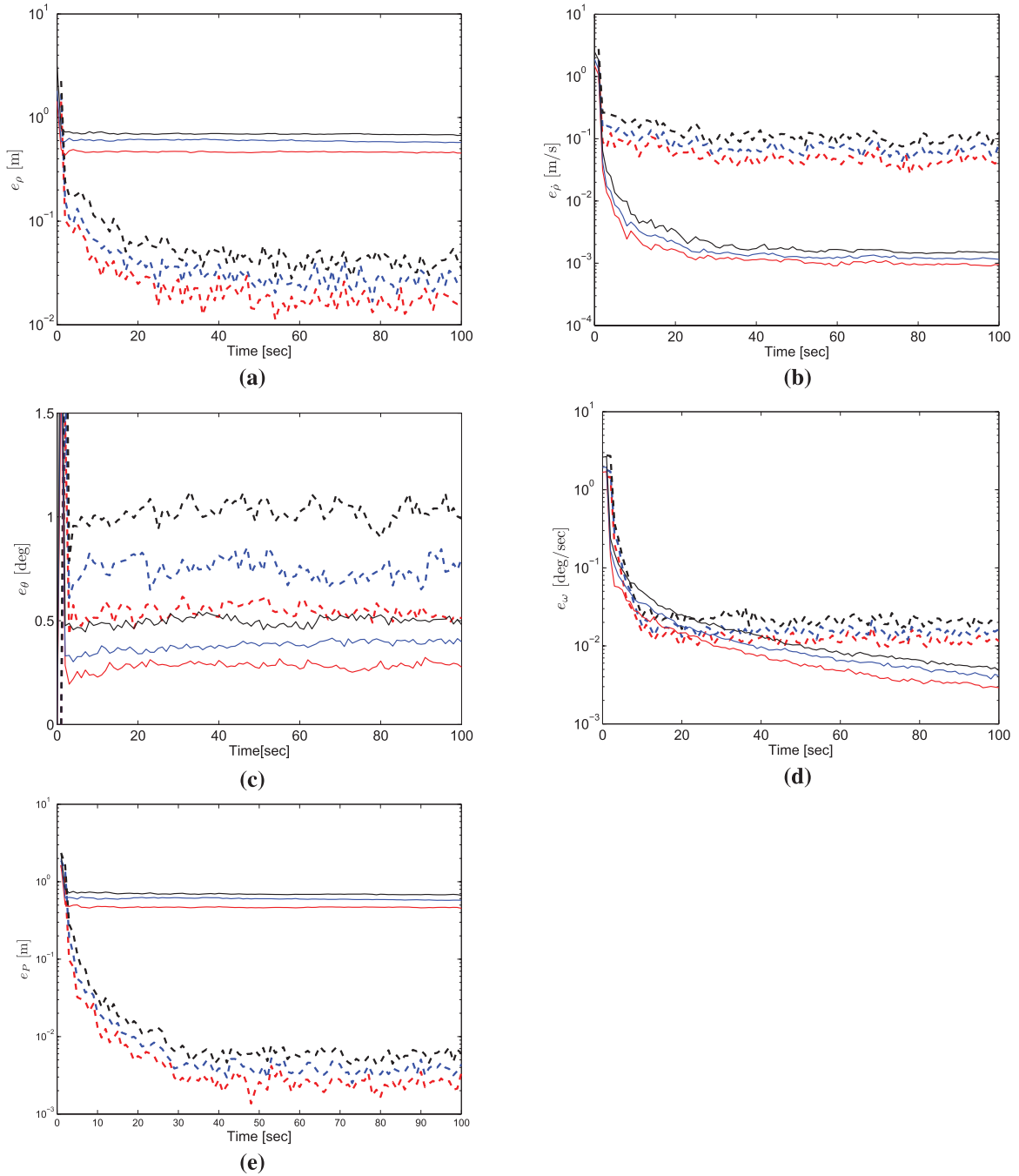


Fig. 4. Effect of kinematic coupling on the estimation errors. The distribution of the estimation errors (shown via the 50, 70, and 90 percentiles) of the plain IEKF filter using either the uncoupled (thin lines) or coupled (thick dashed lines) models. (a) Relative position. (b) Relative translational velocity. (c) Relative attitude. (d) Relative angular velocity. (e) Feature points.

remains almost unaffected. The most vulnerable state is the relative attitude, with the relative attitude estimation error increasing from 0.5° to about 2.25° when the measurement noise covariance is increased by an order of magnitude. Overall, it is safe to state that the algorithm is robust to measurement noise covariance variations.

VI. EXPERIMENTAL VALIDATION

Laboratory experiments for relative spacecraft navigation require simulating, at least to some extent, the dynamics of orbital motion. Laboratories for simulating satellite

orbital dynamics exist at MIT (Space Systems Lab), Stanford (Space Systems Development Laboratory), Caltech (Multivehicle Wireless Test-bed), and the Naval Postgraduate School (Spacecraft Robotics Laboratory). In all of the aforementioned labs, the test-bed comprises a flat-floor or a granite-table facility in which frictionless motion is achieved by releasing compressed air from an on-board pressurized air tanks. In the Naval Postgraduate School, for example, laboratory experiments are conducted for examining docking of a chaser to a collaborative target [38] and for testing multiple spacecraft control algorithms [39].

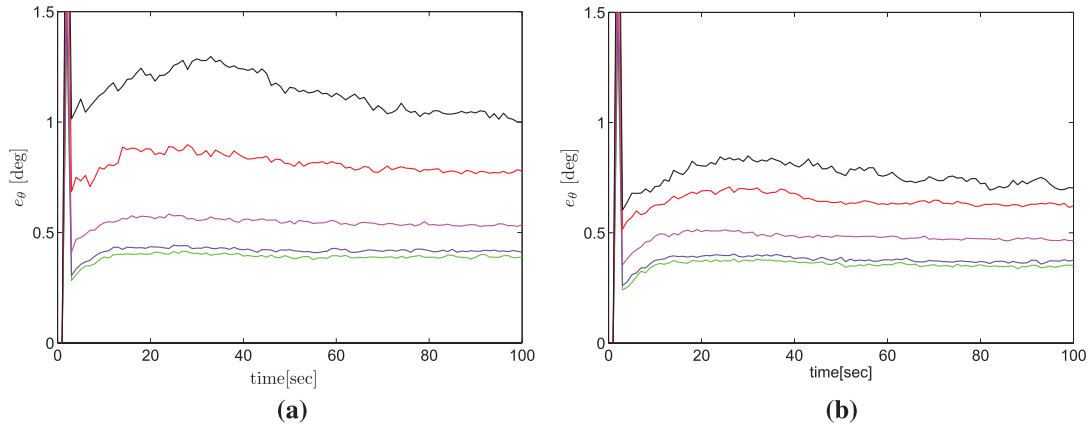


Fig. 5. Distribution of the relative attitude estimation errors (shown via the 45, 50, 70, 90, and 95 percentiles) of the robust multiple IEKFs filtering scheme and of a single IEKF with a randomly chosen target inertia. (a) Without inertia identification. (b) Robust Multiple IEKFs.

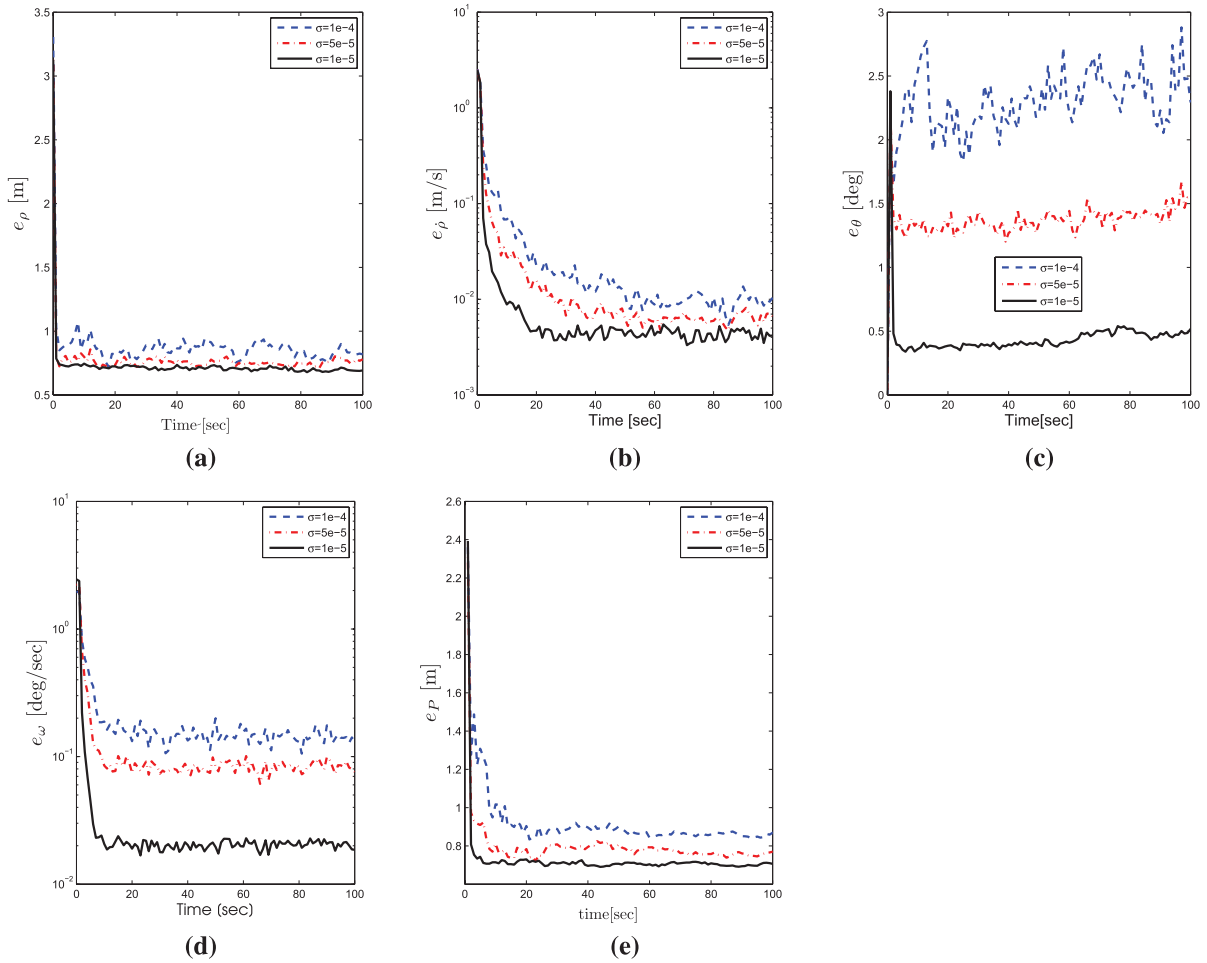


Fig. 6. Effect of varying the measurement noise covariance on the estimation errors. The distribution of the estimation errors (shown via the 90th percentile) of the plain IEKF filter using the uncoupled model. (a) Relative position. (b) Relative translational velocity. (c) Relative attitude. (d) Relative angular velocity. (e) Feature points.

The experiments for evaluating the performance of the stereovision-based relative state estimation algorithm are conducted at the Technion's Distributed Space Systems Laboratory. This lab comprises an air-bearing table and satellite models floating on the table. As opposed to the aforementioned labs, the satellites do not carry their own compressed air

supply. Flotation is achieved by releasing the compressed air through small nozzles constituting a large porous plate. Each satellite is equipped with a rate gyro, an accelerometer, and a three-axis magnetometer. An overhead CCD camera provides the position, heading, and roll angle of the satellite models. The signals generated by the satellite sensors are communi-

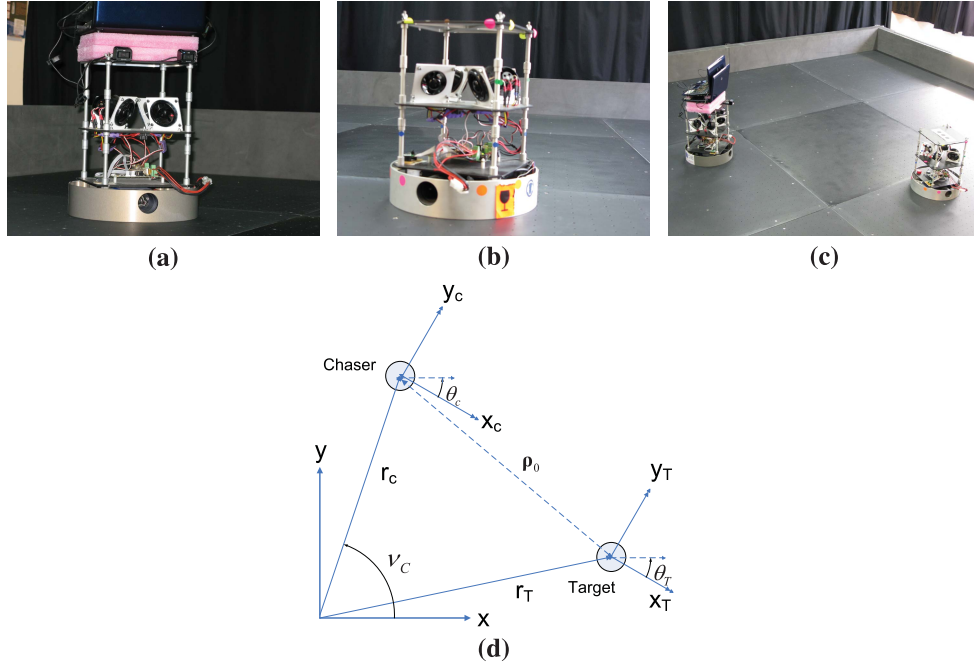


Fig. 7. Upper panel: satellite models. Bottom panel: underlying reference frames. (a) Chaser. (b) Target. (c) Chaser-target setup. (d) Reference and body frames.

cated to a PC using the ZigBee communication protocol. The PC uses these signals as feedback to command the actuators. The system is capable of simulating orbital motion—for example, leader-follower formations—by controlling the desirable relative position and attitude.

A. Experiment Setup

Two satellite models, chaser and target, participate in the experiment. Both satellites are controlled to move along a circular reference orbit in a coorbital formation. The chaser satellite model is equipped with two Microsoft Life-Cam HD-5000 1280 × 720 pixel webcams separated by a 14.5-cm baseline [Fig. 7(a)]. These cameras acquire images of a target satellite model [Fig. 7(b)] with a frame rate of 30 Hz. The chaser–target relative setup [Fig. 7(c)] is modeled using standard reference frames [Fig. 7(d)].

B. Modeling Orbital Dynamics in the Experiment

Based on the lab test-bed, a 2-D orbital motion is simulated. The following state vector is used for modeling the relative dynamics between the chaser and target satellite models: $\mathbf{x} = [\rho_0^T, \dot{\rho}_0, \theta, \omega, P_1, \dots, P_N]$, where $\rho_0 \triangleq \mathbf{r}_T - \mathbf{r}_C = [\rho^{(1)}, \rho^{(2)}]$ is the relative 2-D position between the center of rotation of each satellite, $\theta \triangleq \theta_T - \theta_C$ is the relative body angle (relative azimuth) and $\omega = \dot{\theta}$ is the magnitude of the relative angular velocity.

To implement the relative satellite translational-dynamics equation (5) in the experiment, a scale factor for the gravitational constant is introduced. The control system in the experiment steers both the chaser and the target to move on a nominal circular orbit with orbital angular velocity $\dot{v}_C$.

The centrifugal acceleration can be expressed as

$$\mathbf{a} = \dot{v}_C^2 r_C \hat{\mathbf{r}}_C. \quad (65)$$

On the other hand, for a nominal orbit under Newtonian gravity

$$\mathbf{g} = \frac{\mu}{r_C^2} \hat{\mathbf{r}}_C. \quad (66)$$

Thus, we can define the scale factor μ^* , playing the role of the gravitational parameter, as

$$\mu^* \approx \dot{v}_C^2 r_C^3. \quad (67)$$

Using this scale factor, the relative translational dynamical equation for the 2-D experiment can be written as follows:

$$\ddot{\rho}_0^{(1)} = 2\dot{v}_C \dot{\rho}_0^{(1)} + \dot{v}_C^2 \rho_0^{(1)} - \frac{\mu^* (r_C + \rho_0^{(1)})}{\left[(r_C + \rho_0^{(1)})^2 + (\rho_0^{(2)})^2 \right]^{3/2}} + \frac{\mu^*}{r_C^2} \quad (68)$$

$$\ddot{\rho}_0^{(2)} = -2\dot{v}_C \dot{\rho}_0^{(1)} + \dot{v}_C^2 \rho_0^{(2)} - \frac{\mu^* \rho_0^{(2)}}{\left[(r_C + \rho_0^{(1)})^2 + (\rho_0^{(2)})^2 \right]^{3/2}} \quad (69)$$

where v_C is the chaser polar angle and r_C is the chaser CM radial distance to the (virtual) center of attraction. In the experiments the value of the angular velocity $\dot{v}_C = 0.0524$ rad/s and $r_C = 1$ m, so from (67) $\mu^* \approx 0.0027$ m³/s². With N_C and N_T denoting the external torque magnitudes acting on the chaser and target, respectively [see (6)], the relative rotational dynamics are expressed as follows:

$$\dot{\omega} = N_T \frac{1}{I_T} - N_C \frac{1}{I_C} \quad (70)$$

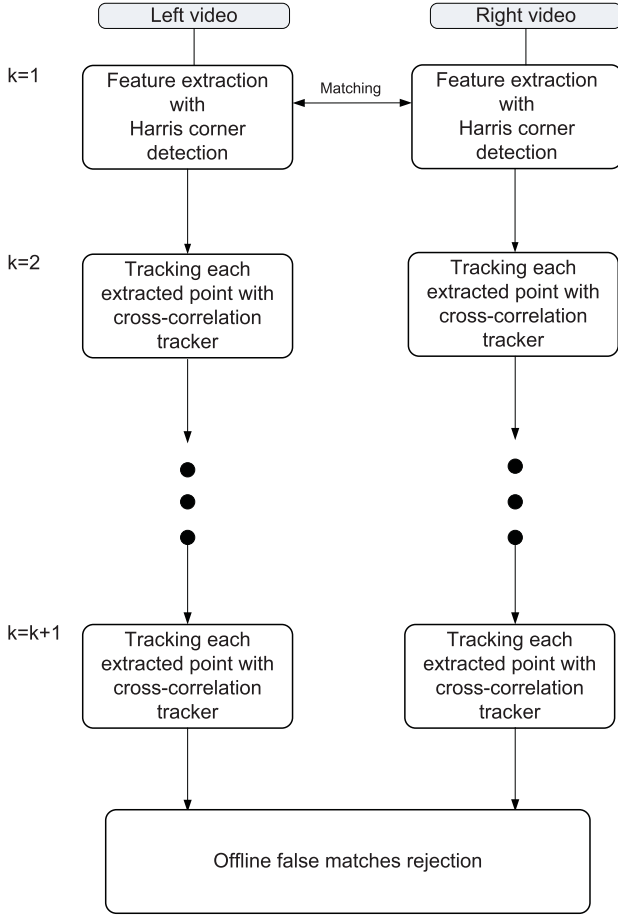


Fig. 8. Flowchart of the observation processing in the experiment.

where I_T and I_C are the target and chaser moments of inertia about the principal axis of minimum inertia. In the experiment, the control torques were used to point the models to the (virtual) center of attraction. Therefore, the applied control torques were similar, so that

$$\dot{\omega} \approx N_C \left(\frac{1}{I_T} - \frac{1}{I_C} \right). \quad (71)$$

C. Observation Processing

The estimation algorithm is based on the assumption that a constant set of N feature points on the target is tracked by both the right and left cameras. Implementing a stereovision feature points tracker in the experiment is carried out in three stages: Features extraction, feature matching, and feature tracking, as shown in the flowchart in Fig. 8.

1) *Feature Extraction*: First, the left and right videos are synchronized based on a discrete event observed by both cameras. Then, a set of features is extracted independently in the first (synchronized) left and right frames using a Harris corner detector [40].

2) *Feature Matching*: As the camera baseline is relatively small, the same set of features is likely to be extracted in the left and right cameras. To match these two sets, a matching algorithm is used. This algorithm is searching for correlated points within given windows surrounding each

point. The set of matches contains the feature points that are maximally intercorrelated in the left-right and right-left directions.

3) *Feature Tracking*: The third stage deals with tracking the set of the left and right matched feature points. As the cameras are working in high frequency (30 Hz), the interframe motion is relatively small. Therefore, we used a normalized cross-correlation-based tracker to track the same group of the selected feature points in the left and right video movies. The normalized cross-correlation between two small windows w_1 and w_k is defined by

$$\xi(w_1, w_k) = \frac{\sum_{i,j} (w_1(i, j) - (\bar{w}_1)) \cdot (w_k(i, j) - (\bar{w}_k))}{\sqrt{\sum (w_1(i, j) - \bar{w}_1)^2 \cdot \sum (w_k(i, j) - \bar{w}_k)^2}} \quad (72)$$

where $\bar{w}_1$ and $\bar{w}_k$ are the mean values of the intensity in these windows.

Starting from the selected set of features at the time point $k = 1$, the cross-correlation tracking algorithm searches at every time point k , within a prescribed search radius, the window w_k that maximizes the correlation with the window w_1 . The coordinates of a feature point in a successive frame are determined as the center of the window that maximizes the correlation function in (72). This algorithm is acting upon each one of the selected feature points. A typical image of the tracked feature points in the left and right cameras is shown in Fig. 9.

Ideally, the two cameras are separated only horizontally. Thus, the vertical coordinate of any projected feature point is assumed to be the same for the left and right cameras. A vertical disparity may exist only when there is a misalignment between the separating baseline and the horizontal line. In any case, however, this vertical disparity would be time-invariant. Based on this fact, false matches can be rejected. Finally, the set of matched points successfully tracked over the movie frame sequence is used for generating the observation vector for the relative motion filter described in the previous sections.

D. Experiment Results

The experiment results are shown in Figs. 10–12. Fig. 10(a) shows the estimation of the range between the chaser and the target's center of rotation, whereas Fig. 10(b) shows the feature point estimation error as defined in (64) (to determine the ground truth of the feature points location, we manually measured the location of the feature points on the target after the off-line processing). It is seen that the range is estimated with an accuracy better than 20 cm, and that the structure is recovered to an accuracy of about 5 cm. Fig. 11 shows the estimation performance of the relative azimuth and relative angular velocity. The upper plots [Fig. 11(a) and (b)] show the real and estimated signals. The lower plots [Fig. 11(c) and (d)] show the estimation errors and the IEKF-computed 1σ and 3σ envelopes. The steady-state estimation error in the relative azimuth is about 0.5° . Evidently, the estimation of the angular velocity is unbiased.

In this experiment, the inertia identification algorithm using multiple IEKFs is also tested. Finally, five hypotheses of the

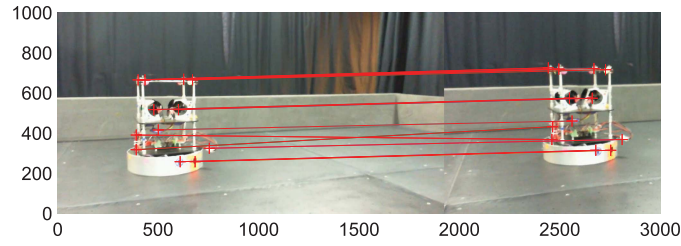


Fig. 9. Left and right feature point matches: the set of matched feature points is selected at the first frame and then the features are tracked over time for each camera separately using a cross-correlation tracker.

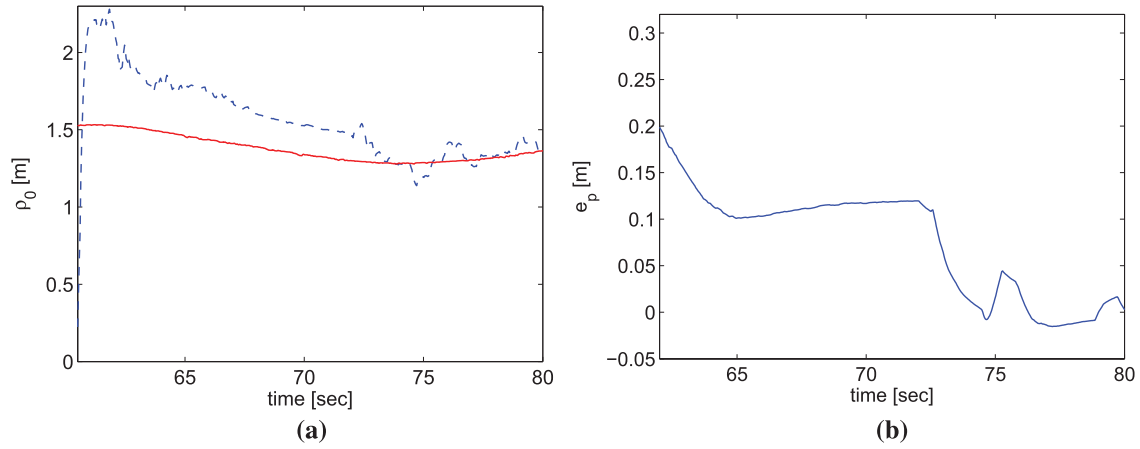


Fig. 10. (a) Actual (solid line) and estimated (dashed line) range. (b) Estimation errors of the feature points.

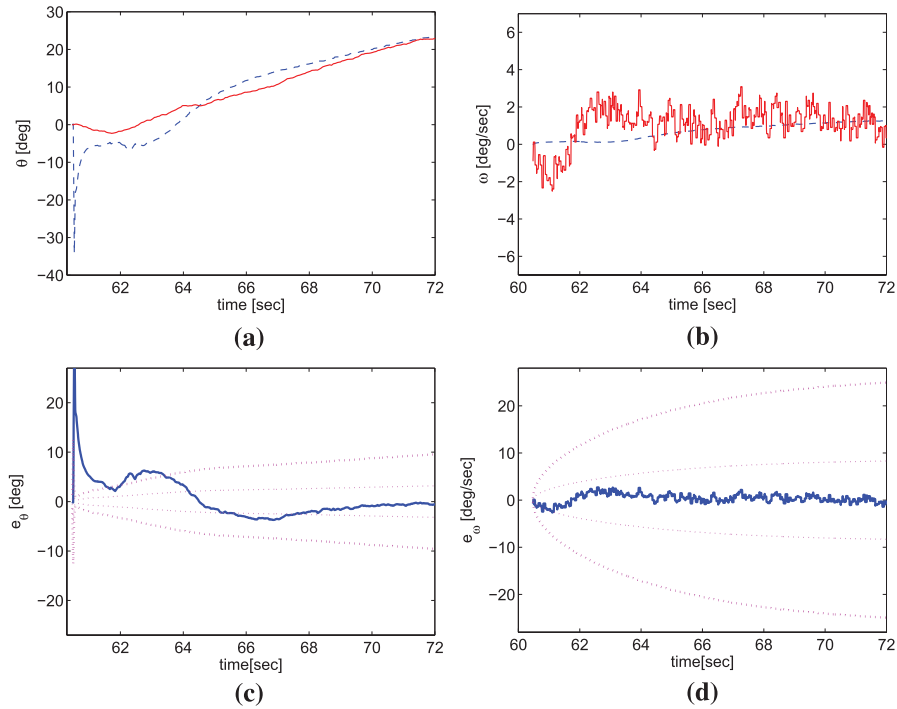


Fig. 11. Estimation performance in the azimuth and angular velocity channels. Upper panel: actual (solid line) and estimated (dashed lines) signals. Bottom panel: estimation errors (thick solid lines) with the IEKF-computed (thin and thick dotted lines) 1σ and 3σ envelopes. (a) Azimuth. (b) Angular velocity. (c) Azimuth errors. (d) Angular velocity errors.

target inertia sampled randomly from the uniform distribution $I_T \sim \mathcal{U}(0.05 \text{ kg m}^2, 0.2 \text{ kg m}^2)$, where the real target inertia is 0.1 kg m^2 . Fig. 12 shows the inertia estimation process and its effects on the angular velocity estimation. Fig. 12(a) shows

that the moment of inertia is identified to within an accuracy of 10%. Following the inertia identification, the estimated relative angular velocity converges to the actual value, as seen in Fig. 12(b), and the the IEKF-computed 1σ and 3σ envelopes

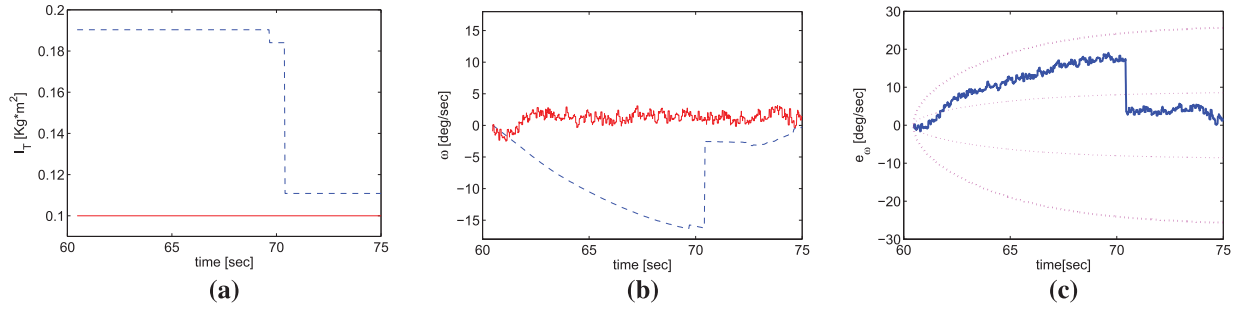


Fig. 12. Performance of the filtering algorithm with the inertia identification scheme. Left: actual (solid line) and estimated (dashed lines) signals. Bottom panel: estimation errors (thick solid lines) with the IEKF-computed (thin and thick dotted lines) 1σ and 3σ envelopes. (a) Inertia. (b) Angular velocity. (c) Angular velocity errors.

of the relative angular velocity estimation error, shown in Fig. 12(c), exhibit consistent behavior.

These results verify the previous treatment, showing that the relative state including the position, angle and angular velocity can be estimated. The estimation algorithm also identifies the closest moment of inertia of the target from several options, thereby reducing the estimation error in the relative angular velocity, and is capable of reducing the uncertainty in the target's structure.

E. Scalability and On-Orbit Implementation

The above reported experiment indicates that a stereovision system employing the algorithms developed in this paper could be implemented on board operational satellites. The main question to be asked is how the main characteristics of the sensors (field of view, baseline distance, and resolution) would scale up to orbital distances and actual lighting conditions in orbit. As relative distances between the satellites increase, the baseline distance between the cameras and the optical aperture should also be increased. This implies that in real applications, stereovision systems may be limited to distances of a few kilometers, depending on environmental factors. A thorough discussion of these aspects may be found elsewhere [41].

VII. CONCLUSION

This paper developed a stereovision-based filtering algorithm for estimating the relative state between two noncooperative spacecraft. The proposed scheme employed multiple IEKFs, each endowed with a different hypothetical target inertia. This architecture was shown to significantly improve the robustness of the algorithm to inertia modeling uncertainties. In addition, this paper corroborated the benefits of incorporating kinematic coupling between the translational and rotational dynamics: MC simulations showed that about 2-cm accuracy (50th percentile) can be obtained in range estimation. The theoretical analysis showed, through the concept of estimability, that attitude and structure information was contained in the visual measurements, and that it can be used for estimating the complete relative state. The validity of the theoretical results was demonstrated experimentally. This experiment supported the numerical simulations, showing that a structure estimation error smaller than 5 cm can be obtained.

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